

YIKE SHI

yikeshi9248@ucla.edu | 310 706 5575 | linkedin.com/in/yike-shi2004 | yik3.github.io

Education

University of California, Los Angeles (UCLA), B.S. Computer Engineering Sept 2023 - Dec 2026

- GPA: **3.95/4.0**, Dean's Honors List $\times 8$, Magna Cum Laude
- HKN-IEEE Honor Society Workshops Chair, Samueli Research Scholar(2025)
- John DeGross Haller Memorial **Merit** Scholarship, Kalosworks Centennial ECE **Merit** Scholarship
- Coursework: Robotics, Deep Learning, Machine Learning, Computer Vision, Feedback Control, Computational Imaging (Graduate-level), Algorithm, Software Construction, IoT, Reinforcement Learning, OS and Architecture

Experience

Robotics Software Engineer, HawkRobo June 2026 – Now

- Architecting modular ROS2/Python control software and data pipelines, decoupling hardware communication from policy execution to improve system reliability during robot-dog-open-door deployments
- Built a quadruped mobile manipulation platform with a robotic arm on real hardware and MuJoCo simulation
- Executed large-scale Reinforcement Learning (RL) in MuJoCo, leveraging extensive domain randomization (varying door geometry, kinematics, and spatial positioning) to train robust control policies

Robotics Researcher, UCLA Robot Intelligence Lab (URIL) Jan 2025 – Now

- Assembled the URIL ACT ALOHA **bimanual teleoperation** hardware system. Created a ROS+OpenCV-based 50 Hz real-time data pipeline for arm and camera data, and a set of Python APIs that calls ROS for inference
- Built and designed the python functions for Realman Humanoid Robot Arm that control arms and fingers

Learning where to look: Gaze-Guided Active Perception for Long-Horizon Imitation Learning

- Spearheaded the development of a gaze-conditioned policy to resolve perceptual ambiguity in long-horizon tasks; designed a pipeline translating human visual intent to robot actions using Meta Aria glasses
- Architected a probabilistic gaze prediction model integrating MDN, GMM, and UNet to infer sub-goal intent, enabling dynamic decomposition of complex tasks into executable skills
- Engineered a gaze-centric skill selector using LSTM and a foveated vision model (simulating attention via Gaussian noise injection), achieving a **16% increase** in task success rate over SOTA baselines
- Presented research on Meta's Research Summit for Egocentric Perception 2025 and UCLA Research Symposium

DexEXO: A Wearability-First Dexterous Exoskeleton for Operator-Agnostic Demonstration and Learning

- Engineered an edge data ingestion framework across multimodal sensors (Cameras, IMU, LiDAR, Encoders), ensuring 50Hz hardware time synchronization during high-frequency collection
- Developed real-time telemetry monitoring and automated post-processing pipelines (low-pass filtering, lever-arm compensation) to validate, clean, and convert raw data stream into ML-ready dataset formats
- Standardized data collection workflows and operator diagnostic tools, accelerating the acquisition of 700+ high-quality demonstrations used for end-to-end UNet Diffusion Policy training. Achieved **90%** success rate
- Key Contributions: A much more simplified learning pipeline with no inpainting and replay through negligible visual gap; Achieved SOTA performance without tactile sensors; Improved wearability from user studies

Learning from One-Shot Human Video for Cross-Embodiment Optimization and Reasoning

- Developed a one-shot video-to-policy pipeline enabling a Franka arm to execute complex tasks from a single, unstructured human demonstration without dense motion retargeting
- Formulated automated spatial & physical constraint reasoning using **VLMs** to generate near-optimal, code-based execution programs while bypassing manual prompt engineering
- Built and validated custom MuJoCo simulation environments, achieving an average execution reward of 0.91/1.00 and an 89% task success rate on standard manipulation benchmarks

Publications

Zhu, A., ..., **Shi, Y.**, Wu, Y., ... Cui, Y. & Hong, D. W. (2026). DexEXO: A Wearability-First Dexterous Exoskeleton for Operator-Agnostic Demonstration and Learning. arXiv preprint arXiv:2603.17323. **IROS 2026 Accepted, ICRA 2026 Data to Decisions Workshop Oral (top 5%)**

Zhu, M., ..., **Shi, Y.**, ... Cui, Y. & Hong, D. W. (2026). WHED: A Wearable Hand Exoskeleton for Natural, High-Quality Demonstration Collection. arXiv preprint arXiv:2602.17908. **IEEE Ubiquitous Robots 2026**

Shi, Y. A robust and novel semantic segmentation deep neural network for robotic surgery vision with a single RGB camera. (ICAICE 2022) SPIE.

Technical Skills

Languages: Python, C++, C, MATLAB, Verilog, SQL, JavaScript, HTML/CSS

Robotics & AI: PyTorch, NumPy, ROS/ROS2, Isaac Sim, MuJoCo, OpenCV, TensorRT, Git/GitHub, Teleop-System

Dev Tools & Cloud: Docker, Linux/Bash, AWS, Node.js, React, CI/CD